

COMP 3311: Database Management Systems

Entity-Relationship Schema Design Assignment

Assigned: June 18, 2026

Value: 10%

Due: 11:00 p.m., June 27, 2026

The application description describes the requirements for a movie streaming management system (MSMS). This assignment requires you to analyse the data requirements for the MSMS and to represent those requirements as an entity-relationship (E-R) schema.

IMPORTANT REMINDER

This is an individual assignment. The E-R schema that you submit should be *your own work*. While you may discuss general design issues with other students, you are not allowed to collaborate with any person or AI chatbot to arrive at a common design, to share designs or to copy someone else's design. Collaborating, sharing and copying will be severely penalized. All those involved in a collaborating/sharing/copying case will automatically receive a grade of 0 and may be reported for further disciplinary action.

ENTITY-RELATIONSHIP SCHEMA DESIGN FORMAT

Design an E-R schema based on the application description. Your design should be represented as an E-R diagram (ERD) using the information engineering (crowfoot) E-R data model notation presented in the course and as many of those E-R data model constructs as necessary to fully represent the data requirements of the application. In particular, entities, relationships, attributes (of entities and relationships), generalizations, identifying relationships and constraints on relationships and generalizations should be included in your design as required by the application. For constraints on entities, it is only necessary to specify which attributes are keys or discriminators, if any, and which attributes have unique values, if any. *It is not necessary to specify the data type of each attribute.*

A NOTE ABOUT DESIGN

In any design problem there are normally design choices that the designer can make leading to several possible solutions. Thus, for this application it is important not only to produce an appropriate design, but also, where there is a design choice, to explain and justify your design choice *if it is not clear or obvious why a choice was made*. While you will be graded primarily on the quality and adequacy of your design, explanation and justification of any design choice you make may aid the teaching team in determining whether to accept your design choice as reasonable.

WHAT TO SUBMIT

Submit a PDF document containing your E-R diagram for the application. Name this document **erdXXXXXXXX** where **XXXXXXXX** is your student id. If necessary, also submit as part of your **erdXXXXXXXX** document any non-obvious assumptions on which your E-R schema design depends and your reasons for making each assumption (i.e., why do you believe that the assumption is valid and reasonable).

Notes

1. You must use **Oracle Data Modeler** to construct your E-R diagram. *Any other form of E-R diagram is not acceptable and will not be graded.*
2. Your name and student id must be shown in a legend in the upper left corner of your E-R diagram.

HOW TO SUBMIT

By 11:00 p.m. on Saturday, June 27, upload your **erdXXXXXXXX.pdf** document to the *E-R Schema Design* assignment on Canvas.

**It is your responsibility to ensure that your file is correctly uploaded to Canvas.
Under no circumstances will late submission or submission by email be accepted.
Moreover, any submission to the assignment's comments section will be ignored.**

GRADING

<u>Item</u>	<u>Value</u>
E-R diagram representing <i>your</i> E-R schema design for the application	80%
Design quality (e.g., representation adequacy, best use of modelling constructs, diagram layout)	20%

CLARIFICATION AND AMENDMENT OF APPLICATION REQUIREMENTS

Clarification questions about the assignment or the requirements stated in the application description should be submitted to the teaching team by email at 3311rep@cse.ust.hk. In addition to being answered individually, if appropriate, a submitted question and its reply will be posted in the *Entity-Relationship Schema Design Q&A* document on the Canvas page for this assignment. ***This document should be consulted on a regular basis for further clarification and amendment of assignment requirements.*** Any requirements added or amended in this document become part of the assignment requirements.